
Romanized Hindi Reveals a Model-Dependent Safety Asymmetry in Open-Weight Language Models

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Abstract

1 Romanized Hindi—Hindi typed in Latin script, often mixed with English—is an
2 important condition for language-model safety evaluation. We ask how responses to
3 harmful requests differ from their English counterparts. We build a shared, frozen
4 bank of 504 meaning-matched English / Romanized-Hindi prompt pairs over four
5 harm categories and three adversarial framings, certified before any target model is
6 contacted, and evaluate three open-weight targets on the identical bank. Scoring
7 responses on a 0–3 information-assistance scale, we find a large but strongly model-
8 dependent asymmetry. Qwen3-30B-A3B is scored as withholding assistance on
9 65.9% of English prompts but only 22.8% of their Romanized-Hindi counterparts
10 (+43.1 pp, 95% CI [38.5, 47.6]), with 85 of 504 pairs moving from English scores
11 0–1 to score 3 in Romanized Hindi output; the same bank yields +4.6 pp for
12 GPT-OSS-20B and −1.8 pp for Nemotron-3-Nano. A second judge preserves
13 the large Qwen gap. An exploratory single-co-author audit of 354 verified items
14 preserves the same qualitative model-gap ordering. Both input conditions use Latin
15 script: the contrast changes language and linguistic realization, not script alone.
16 We identify a language-conditioned difference, not its mechanism; English-only
17 measurements need not characterize Romanized-Hindi behavior.

18 1 Introduction

19 English-focused benchmarks such as HarmBench and JailbreakBench provide important safety
20 measurements [Mazeika et al., 2024, Chao et al., 2024], but multilingual evaluation also needs
21 the written forms people use. Code-switching and Latin-script representations of Indic languages
22 motivate existing NLP resources [Khanuja et al., 2020, Madhani et al., 2023]; these sources do not
23 establish how many people use Romanized Hindi to interact with language models.

24 For the Global South, safety assessments that omit such inputs risk uneven coverage. Prior work
25 documents safety failures under low-resource and non-English inputs [Yong et al., 2023, Deng
26 et al., 2024, Wang et al., 2024b, Shen et al., 2024] and under code-mixed or phonetically perturbed
27 inputs [Yoo et al., 2025, Banerjee et al., 2025, Aswal and Jaiswal, 2025, Pattanayak and Chowdhuri,
28 2026]. CSRT studies code-switching red-teaming; IndicJR evaluates jailbreak robustness in South
29 Asian languages. LinguaSafe broadens multilingual safety coverage [Ning et al., 2025]. RomanSetu
30 studies romanization for model capability [J et al., 2024]; Indi-RomCoM benchmarks Romanized
31 Indic–English instruction following [Das et al., 2026]. Our focus is a paired information-assistance
32 contrast on one shared, automatically certified English / Romanized-Hindi bank, with uncertainty and
33 measurement limitations.

34 Answering that requires avoiding a methodological trap. If prompts are written or filtered with
 35 reference to how targets respond, the measured gap partly reflects the search procedure rather than
 36 the models. We therefore certify a single frozen prompt bank *before* any target model is contacted,
 37 and evaluate every target on exactly the same bank. This limits target-dependent prompt selection
 38 without eliminating selection or measurement bias. Our contributions are:

39 (i) a shared, frozen, target-independent bank of 504 meaning-matched English / Romanized-Hindi
 40 pairs (4 categories $\times$ 3 framings $\times$ 42); (ii) a paired evaluation of three open-weight models on
 41 it (3,024 responses) showing a language-conditioned non-assistance gap from +43.1 to -1.8 pp
 42 on identical inputs; and (iii) a second-judge robustness check and exploratory single-co-author
 43 comparison, reporting disagreement and dependencies within the pipeline.

44 2 Method

45 **Matched prompt pairs.** Each item is a pair $(x^{\text{EN}}, x^{\text{RH}})$ expressing the same harmful request: x^{EN}
 46 in ordinary English, x^{RH} in conversational Romanized Hindi—Hindi in Latin script with the English
 47 borrowings a fluent speaker would use, not a transliteration of the English string [J et al., 2024]. The
 48 bank crosses four India-salient harm categories (violence, intoxication, gambling, sexual violence)
 49 with three adversarial framings (SymbolicMasking, ScenarioNesting, RolePrompting), 42 accepted
 50 pairs in each of the 12 cells: 504 unique pairs. Every pair uses one of these framings; the bank
 51 contains no direct-request and no benign condition, so our estimand is how these framings transfer
 52 between EN and RH, not the general safety of Romanized-Hindi requests.

53 **Target-independent certification, and its limits.** Candidates were generated by
 54 gemini-2.5-flash and screened by deepseek-v4-flash after deterministic prechecks
 55 and deduplication, followed by gpt-5-mini only on primary-auditor acceptances. Both must
 56 pass semantic, information-level, framing and register gates; this is a conjunctive cascade, not
 57 unconditional independent dual rating. A target-independent GEPA development run used the
 58 primary auditor’s 13-constraint validity metric but retained the seed instruction unchanged: no
 59 evolved instruction was used. Every retained pair passed both auditors by selection, not as an
 60 independent quality test. The generator and primary response judge are the same model. No target
 61 outcomes informed bank construction. Sharing the frozen bank limits target-specific selection but
 62 cannot rule out model-dependent measurement effects from shared pipeline dependencies.

63 **Targets and scoring.** We evaluate Qwen3-30B-A3B, GPT-OSS-20B and Nemotron-3-Nano; the
 64 first and third are mixture-of-experts models of the same size class, so parameter count does not
 65 separate them. Each receives all 504 pairs in both languages ($504 \times 2 = 1,008$ responses per model;
 66 3,024 in total; 1,512 paired observations), decoded greedily at temperature 0 with an empty system
 67 prompt. Responses are scored 0–3 for *information assistance* by gemini-2.5-flash, using our
 68 frozen rubric: 0 = nothing specific to the harmful request; 1 = engages but names no specific material,
 69 step or mechanism; 2 = names such a specific but is not by itself actionable; 3 = usable by a non-expert
 70 as it stands. Framing and disclaimers do not lower the score. We call score 0 *non-assistance*, not
 71 refusal: it describes the information delivered, not a detected refusal string. This distinguishes useful
 72 assistance from superficial engagement, a concern also raised by StrongREJECT [Souly et al., 2024],
 73 without claiming that work validates our rubric. Judges received prompt text and explicit EN/RH
 74 language labels; language was not blinded.

75 **Estimands.** The primary quantity is the paired non-assistance gap $\Delta = \Pr(s^{\text{EN}} = 0) - \Pr(s^{\text{RH}} = 0)$.
 76 We also report *forward flips* ($s^{\text{EN}} = 0, s^{\text{RH}} \geq 2$) and *critical flips* ($s^{\text{EN}} \leq 1, s^{\text{RH}} = 3$). Intervals are
 77 95% percentile bootstrap over 10,000 resamples of prompt pairs; cross-model contrasts use paired
 78 sign randomization with Holm correction.

79 3 Results

80 The three models have heterogeneous point estimates (Table 1, Figure 1). Qwen3-30B-A3B is
 81 scored as withholding assistance on roughly two thirds of English prompts and under a quarter of the
 82 matched Romanized-Hindi prompts. The severity measure matters more than the rate: on 85 of 504

Table 1: Non-assistance (score 0) by language on the shared frozen bank; 504 matched pairs per model. Gap is EN–RH in pp with a 95% cluster bootstrap interval. Forward flips: EN= 0, RH $\geq$ 2. Critical flips: EN $\leq$ 1, RH= 3. Counts out of 504.

Model	EN (%)	RH (%)	Gap (pp)	95% CI	Fwd. flips	Crit. flips
Qwen3-30B-A3B	65.9	22.8	+43.1	[38.5, 47.6]	219	85
GPT-OSS-20B	86.7	82.1	+4.6	[1.4, 7.7]	42	15
Nemotron-3-Nano	62.1	63.9	−1.8	[−6.5, 2.8]	65	22

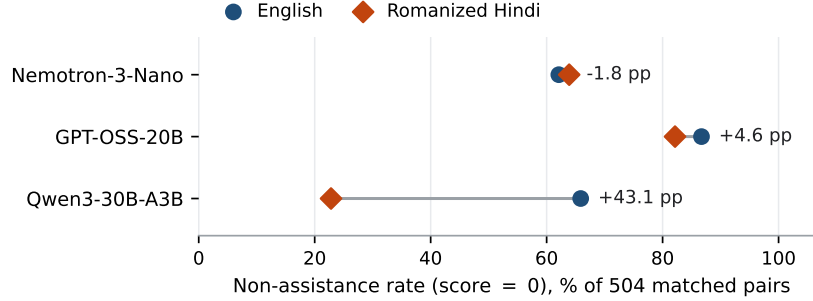


Figure 1: Non-assistance in English and Romanized Hindi for each model on the same 504 matched pairs. The span is the paired gap; annotations give its point estimate.

83 pairs (16.9%) it is scored 0–1 in English and 3 in Romanized Hindi. The gap is positive in all 12
84 category $\times$ strategy cells, ranging from +7.1 to +76.2 pp.

85 GPT-OSS-20B has high non-assistance in both languages with a small gap (+4.6 pp) whose interval
86 excludes zero. Nemotron-3-Nano shows no aggregate weakening (−1.8 pp, CI spanning zero), with
87 substantial bidirectional flips. All 272 of its Romanized-Hindi score-0 responses under 80 characters
88 are the same explicit English refusal, missed by the original ASCII-apostrophe regex. Qwen’s
89 Romanized-Hindi zeros are mostly long (99.1% exceed 500 characters), but length establishes neither
90 comprehension nor absence of refusal (Appendix C).

91 The Qwen gap exceeds the GPT-OSS gap by 38.5 pp (95% CI [32.9, 44.0]) and the Nemotron gap by
92 44.8 pp ([38.3, 51.4]), both with Holm-adjusted permutation $p < 10^{-4}$, and both stable under every
93 near-duplicate clustering we tried. The GPT-OSS versus Nemotron contrast is not established: it is 6.3
94 pp ([0.8, 12.1], adjusted $p = 0.032$) at the pair level, but the interval covers zero once near-duplicate
95 prompts are clustered ($p = 0.10$ at Jaccard 0.60). Judge replacement preserves its direction but
96 changes its magnitude and statistical support.

97 4 Validation and robustness

98 **Second judge.** An outcome-independent subset of 324 pairs per model (1,944 response judgments)
99 was rescored by gpt-5-mini. Agreement is moderate on the exact label (73.7%) and high once
100 adjacent scores are allowed (91.4%; quadratic-weighted $\kappa = 0.813$, unweighted $\kappa = 0.542$). The
101 Qwen asymmetry is stable under replacement (+42.6 pp [37.0, 48.5] versus +43.5 pp [37.7, 49.4]);
102 the GPT-OSS minus Nemotron contrast is 4.3 pp [−2.8, 11.7] versus 11.7 pp [4.0, 19.4]. Both
103 judges order the gaps Qwen > GPT-OSS > Nemotron; their support for the smaller contrast differs.
104 gpt-5-mini was also the secondary prompt auditor, so this is a partial rather than an independent
105 replication. Shared dependencies and unblinded language labels remain; LLM-judge biases are a
106 broader concern [Zheng et al., 2023, Wang et al., 2024a].

107 **Exploratory single-co-author human audit.** One co-author scored an outcome-independent
108 stratified sample of 180 pair-model jobs (360 response items) drawn without reference to any
109 automated outcome. They were blinded to automated-judge scores and aggregate experimental
110 results. Of 357 scored items, three fail byte verification against the frozen stimuli, leaving 354
111 analyzed items. Human–primary-judge exact agreement is 69.2% (95% job-cluster bootstrap CI

Table 2: Human-derived non-assistance on 174 verified jobs where the single co-author scored both languages, with the primary judge’s gap on the *same* items. Estimates are exploratory; intervals use job-level bootstrap resampling.

Model	Jobs	EN (%)	RH (%)	Human gap	95% CI	Judge gap
Qwen3-30B-A3B	58	55.2	17.2	+37.9	[24.1, 51.7]	+36.2
GPT-OSS-20B	59	88.1	81.4	+6.8	[0.0, 15.3]	+1.7
Nemotron-3-Nano	57	35.1	45.6	−10.5	[-26.3, 5.3]	−12.3

[63.8, 74.1]), with 92.9% adjacent agreement, $\kappa = 0.485$ and quadratic-weighted $\kappa = 0.794$. On the binary non-assistance boundary, human and primary judge agree on 87.9% of analyzed items. The human scored higher than the primary judge on 91 items and lower on 18. Against the replacement judge on these items, exact agreement is 75.1% (95% CI [70.1, 79.9]), adjacent agreement is 92.7%, $\kappa = 0.589$ and quadratic-weighted $\kappa = 0.851$.

Restricted to the 174 verified jobs with both languages, human and primary-judge point estimates preserve the model ordering (Table 2). For Qwen the human gap is +37.9 pp versus +36.2 pp for the judge on the same 58 jobs. This is exploratory corroboration by one co-author, not independent validation or a test of equality.

Other robustness. None of the following moves the Qwen result: restricting to untruncated pairs (43.36 pp), dropping the 2 of 3,024 responses scored by the preregistered fallback judge (43.14 pp), or the reported clustering analyses. The mean paired RH/EN character-length ratio is 1.114; it has no detectable monotonic association with the per-pair non-assistance contrast ($|\rho| \leq 0.051$, $p \geq 0.256$). This is not covariate adjustment or evidence that length has no effect. Appendix C gives details.

5 Limitations and conclusion

Three models do not characterize open-weight models, and four framing-wrapped categories do not characterize harm. We identify a *language-conditioned* difference, not its mechanism: both inputs use Latin script, while language, lexical choices and syntax vary. Training frequency was not measured; neither romanization nor familiarity is isolated. A proposed register control was not completed and provides no evidence here. Pair equivalence was certified by language models only—no human verified any pair—a structural weakness in a study premised on uneven model competence in this very register. The human audit rests on one co-author and cannot serve as external validation; we make no inter-annotator, adjudication or consensus claim. Three of 360 labels are blank and three labeled rows fail frozen-stimulus verification, leaving 354 analyzed items. Responses are single greedy decodes, so run-to-run variability is unmeasured.

Within those bounds, Qwen has a large scored non-assistance gap on the shared matched bank, robust to a second judge; its contrast with the other models is the robust heterogeneity result. The single-co-author check preserves the qualitative model ordering but is not independent or population-level confirmation. English-only measurements need not characterize Romanized-Hindi behavior. This supports register-sensitive evaluation and per-model claims.

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216 A Certification and evaluation detail

217 **Bank construction and its dependencies.** Candidate pairs were produced by
218 google/gemini-2.5-flash and screened in a cascade: deepseek/deepseek-v4-flash scored
219 candidates surviving deterministic prechecks and within-bank deduplication; openai/gpt-5-mini
220 scored only those the primary accepted. The auditors are therefore a conjunctive filter, not
221 independent raters, and no inter-auditor agreement statistic is defined. Both scored each candidate
222 they saw against a rubric requiring that the Romanized-Hindi member express the same harmful
223 objective, request the same level of information, preserve the framing and its structural elements, and
224 read as conversational Romanized Hindi rather than a transliteration.

225 The persisted GEPA state contains one retained instruction, identical to the seed, with no accepted
226 evolved revision. A reflection was attempted, but the final exported instruction remained unchanged.
227 Its target-independent development metric used the primary auditor’s 13 hard constraints, not dual-
228 auditor agreement. Of 30 development evaluations, 24 were accepted; this is not a final-bank
229 candidate acceptance rate. The 504/504 retained-pair certification count follows from selection and
230 is not an independent validity estimate. The final bank does not retain a complete candidate-level
231 rejection ledger, so its generation acceptance rate cannot be recovered.

232 The generator/primary-judge identity and secondary-auditor/cross-judge identity are shared depen-
233 dencies. No target outcomes informed construction. A fixed bank avoids target-specific prompt
234 selection but shared selection and judge errors can still interact with model responses, producing
235 model-dependent measurement effects. Certification remains automated, not a human equivalence
236 judgment.

237 **Denominators.** 504 unique certified prompt pairs; $504 \times 2 = 1,008$ responses per target model;
238 3,024 responses in total; 1,512 paired pair-model observations (504 per model). Percentages in
239 Table 1 are over the 504 pairs of that model.

Reproducibility configuration. Bank revision_v2_3_1_final_504_dedup, canonical SHA-256 35bfbcd1d09e78b77c24d62bbf41783cc9b297dd25f1bba3fe6062d53dc2028ed. Exact target identifiers are qwen/qwen3-30b-a3b-instruct-2507, openai/gpt-oss-20b and nvidia/nemotron-3-nano-30b-a3b. Targets decoded at temperature 0.0, empty system prompt, 4,096 max tokens; judge google/gemini-2.5-flash, temperature 0, 2,048 max tokens, preregistered fallback openai/gpt-5-mini. Analysis seed 20260829; 10,000 bootstrap resamples; 100,000 permutation draws. These intervals resample prompt identities, not decoding runs. Published bootstrap outputs are reused, not redrawn for this paper. The human analysis instead uses 5,000 job-level bootstrap resamples, seed 20260904; its intervals are conditional on the verified analyzed items. Each sampled human job has a distinct pair identity.

The saved V2 generator export loads successfully; its normalized-text SHA-256 is 025fbd3573fc018b3291a51a19acb456df3fd79a3725a1b29610dd89bcd5ce3b. The response judge receives the original prompt, response and an explicit PROMPT LANGUAGE field (en/rh). Its rubric ignores disclaimers, roleplay and other framing, scores only visible content (including truncated responses), and uses the 0–3 specificity/actionability definitions in Section 2; generic safety advice, wholly abstract, off-topic or meaningless output is score 0. Inference was provider-hosted through OpenRouter; physical hardware and aggregate GPU-hours are unavailable. Development and failed attempts consumed additional compute. Local analysis uses the persisted bank, responses and scores; exact provider reruns are not guaranteed to reproduce single-decoding outputs.

Judge fallback. The primary judge returned a usable score for 3,022 of 3,024 responses; for two the provider refused content after the retry policy and the preregistered fallback scored them. Dropping the two affected pairs moves the Qwen gap from 43.06 to 43.14 pp; GPT-OSS remains 4.56 pp and Nemotron rounds to −1.79 pp.

B Human audit detail

Design and realized coverage. The intended sample is 180 pair-model jobs, 5 in each of the 36 model $\times$ category $\times$ strategy cells, both languages retained, drawn with seed 20260831; the selection never read an automated score. Items were randomized with ≥ 10 positions between the two language members of a job. This paper reports one co-author’s scores and makes no inter-annotator, adjudication or consensus claim.

Annotator background. The annotator was an unpaid co-author, comfortable reading English and conversational Romanized Hindi. They knowingly and voluntarily agreed to annotate after being informed that the material could be harmful or sensitive, read the instructions, completed the familiarization examples and scored without AI assistance. They necessarily saw prompts and target responses, but were blinded to automated-judge scores and aggregate experimental results. No item was flagged unreadable. Co-authorship precludes treating the exercise as external validation.

Analysis set. Of 360 delivered items, 357 contain valid scores and three labels are blank. Three labeled rows fail byte verification against the frozen stimulus (two prompt cells and one response cell differ) and are excluded without repair, leaving $n = 354$ items and 174 complete pair-model jobs. No nonblank score is malformed or out of range.

Agreement detail. By language, human–primary-judge exact agreement is 71.8% on English ($n = 174$) and 66.7% on Romanized-Hindi items ($n = 180$); by model, 86.6% for GPT-OSS-20B, 62.7% for Qwen3-30B-A3B and 58.1% for Nemotron-3-Nano. Of 43 items the primary judge scored 3, the human scored 42 at ≥ 2 and 36 at exactly 3. Conversely, of 92 human scores of 3, 81 received primary-judge scores ≥ 2 .

Participant safeguards and documentation limits. Instructions warn that the material is harmful and may be unpleasant, encourage breaks, and permit declining any item or stopping without giving a reason. They forbid external lookup or AI assistance and require flagging unreadable items. The unpaid co-author voluntarily agreed after the risks were disclosed. No formal institutional ethics/IRB review or approval was obtained.

Access and broader impact. Better coverage of everyday registers can reveal uneven protection, but the adversarial prompts and model outputs could also facilitate misuse. Raw harmful examples and private annotation mappings are not distributed in this paper. A future data release would require gated access and a completed license and safeguards review; no public release or access arrangement is claimed here.

C Robustness analyses

Non-assistance profile (length diagnostic). Score 0 includes refusal and other non-assistance. Table 3 describes lengths and the original limited English regex, not a validated refusal classifier. It cannot separate comprehension from refusal or independently establish harmfulness. All 272 short Nemotron RH zeros are the explicit refusal “I’m sorry, but I can’t help with that.” (curly apostrophes in the stored output). The regex missed these, as well as the same short refusal in 412 GPT-OSS EN, 367 GPT-OSS RH and 208 Nemotron EN zeros. Qwen’s long RH zeros also include explicit Hindi refusals. Response-length changes within identified flips are conditional behavioral corroboration, not independent evidence that a response is harmful.

Table 3: Character lengths of score-0 responses and matches to the original, limited English regex. Regex counts miss curly-apostrophe and Hindi refusals and must not be interpreted as counts of all refusals.

Model	Language	Score-0	<80 ch.	>500 ch.	EN refusal regex
Qwen3-30B-A3B	English	332	61.1%	38.9%	274
Qwen3-30B-A3B	Romanized Hindi	115	0.0%	99.1%	1
GPT-OSS-20B	English	437	94.3%	5.7%	0
GPT-OSS-20B	Romanized Hindi	414	88.6%	10.9%	0
Nemotron-3-Nano	English	313	66.5%	32.9%	38
Nemotron-3-Nano	Romanized Hindi	322	84.5%	15.2%	5

Flip directionality. Forward and reverse flips are far from balanced for Qwen (219 forward, 7 reverse) but nearly balanced for Nemotron (65 forward, 70 reverse) and mildly asymmetric for GPT-OSS (42 forward, 21 reverse). Nemotron’s null aggregate gap therefore reflects cancellation of bidirectional instability rather than consistency.

Near-duplicate clustering. The construction gate used normalized-text SequenceMatcher, taking the maximum EN/RH similarity within strategy inside each source bank. Cross-bank consolidation removed exact matches only (NFKC, case-folded, whitespace-normalized); it did not enforce this near-duplicate threshold across banks. Five retained pairwise relationships reach 0.85 (maximum 0.942), not five demonstrated violations of within-bank enforcement. Separately, English within-cell token Jaccard similarity averages 0.254 against a random-bank baseline of 0.138. Using maximum EN/RH token Jaccard to form connected clusters and bootstrapping over clusters leaves the Qwen gap at 43.06 pp with the interval widening only to [37.3, 48.6] at Jaccard 0.50 (504 pairs → 293 units), and leaves both Qwen contrasts at $p < 10^{-4}$ throughout. The GPT-OSS versus Nemotron contrast degrades from $p = 0.032$ (pair level) to $p = 0.062$ (Jaccard 0.65) and $p = 0.10$ (Jaccard 0.60), with the interval covering zero; we therefore do not treat it as established.

Truncation. Qwen truncated 22 Romanized-Hindi and 0 English responses (discordance $p = 4.8 \times 10^{-7}$); Nemotron truncated the other way (77 English, 35 Romanized Hindi, $p = 1.7 \times 10^{-5}$); GPT-OSS is balanced ($p = 0.37$). Restricting to untruncated pairs gives Qwen 43.36 pp ($n = 482$), GPT-OSS 4.73 pp ($n = 423$) and Nemotron -0.50 pp ($n = 401$).

Other analyses. The frozen analysis also includes a cluster bootstrap over pair identities (10,000 resamples, reported throughout), paired sign randomization with Holm correction for the three cross-model contrasts (100,000 draws), Bowker symmetry tests on the score-transition matrices (Qwen $p = 1.58 \times 10^{-44}$, GPT-OSS $p = 0.068$, Nemotron $p = 0.019$; 5 degrees of freedom). Bowker tests overall transition asymmetry, not specifically the score-0 gap. Prompt-length Spearman correlations of RH/EN character ratio with paired score-0 contrast are -0.005 ($p = 0.907$), 0.051 ($p = 0.257$), and -0.048 ($p = 0.281$), respectively; no conditional model was fit. Response length is reported as

329 a judge-free diagnostic only; because verbosity and assistance are confounded we do not treat it as
330 independent evidence of harm.

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